

Kurtosis: Definition, Types, and Importance

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What Is Kurtosis?

Kurtosis is a statistical measure used to describe a characteristic of a dataset. When normally distributed data is plotted on a graph, it generally takes the form of a bell. This is called the bell curve. The plotted data that are farthest from the mean of the data usually form the tails on each side of the curve. Kurtosis indicates how much data resides in the tails.

Key Takeaways

- Kurtosis describes the “fatness” of the tails found in probability distributions.
- There are three kurtosis categories: mesokurtic (normal), platykurtic (less than normal), and leptokurtic (more than normal).
- Kurtosis risk is a measurement of how often an investment’s price moves dramatically.
- A curve’s kurtosis characteristic tells you how much kurtosis risk there is for the investment you’re evaluating.

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Understanding Kurtosis

Kurtosis is a measure of the combined weight of a distribution’s tails relative to the center of the distribution curve (the mean). For example, when a set of approximately normal data is graphed via a histogram, it shows a bell peak, with most of the data residing within three standard deviations (plus or minus) of the mean. However, when high kurtosis is present, the tails extend farther than the three standard deviations of the normal bell-curved distribution.

Kurtosis is sometimes confused with a measure of the peakedness of a distribution. However, kurtosis is a measure that describes the shape of a distribution’s tails in relation to its overall shape. A distribution can be sharply peaked with low kurtosis, and a distribution can have a lower peak with high kurtosis. Thus, kurtosis measures “tailedness,” not “peakedness.”

Distributions with a large kurtosis have more tail data than normally distributed data, which appears to bring the tails in toward the mean. Distributions with low kurtosis have fewer tail data, which appears to push the tails of the bell curve away from the mean.

For investors, high kurtosis of the return distribution curve implies that there have been many price fluctuations in the past (positive or negative) away from the average returns for the investment. So, an investor might experience extreme price fluctuations with an

investment with high kurtosis. This phenomenon is known as kurtosis risk.

Formula and Calculation of Kurtosis

Calculating with Spreadsheets

There are several different methods for calculating kurtosis. The simplest way is to use the Excel or Google Sheets formula. For instance, assume you have the following sample data: 4, 5, 6, 3, 4, 5, 6, 7, 5, and 8 residing in cells A1 through A10 on your spreadsheet. The spreadsheets use this formula for calculating kurtosis:¹

$$\frac{(n-1)(n-2)(n-3)}{3(n-1)^2} \times \frac{n(n+1) \times (\sum s_{xi-x}^4 - (n-2)(n-3))}{(n-1)(n-2)(n-3)}$$

However, we'll use the following formula in Google Sheets, which calculates it for us, assuming the data resides in cells A1 through A10:²

=KURT(A1:A10)

The result is a kurtosis of -0.1518, indicating the curve has lighter tails and is platykurtic.

Calculating by Hand

Calculating kurtosis by hand is a lengthy endeavor, and takes several steps to get to the results. We'll use new data points and limit their number to simplify the calculation. The new data points are 27, 13, 17, 57, 113, and 25.

It's important to note that a sample size should be much larger than this; we are using six numbers to reduce the calculation steps. A good rule of thumb is to use 30% of your data for populations under 1,000. For larger populations, you can use 10%.³

First, you need to calculate the mean. Add up the numbers and divide by six to get 42. Next, use the following formulas to calculate two sums: s² (the square of the deviation from the mean) and s⁴ (the square of the deviation from the mean squared). Note that these numbers do not represent standard deviation; they represent the variance of each data point.⁴⁵

$$s^2 = \sum (y_i - \bar{y})^2 \quad s^4 = \sum (y_i - \bar{y})^4 \quad \text{where: } y_i = \text{ith variable of the sample} \quad \bar{y} = \text{Mean of the sample}$$

To get s², use each variable, subtract the mean, and then square the result. Add all of the results together:

$$\begin{aligned} (27-42)^2 &= (-15)^2 = 225 & (13-42)^2 &= (-29)^2 = 841 & (17-42)^2 &= (-25)^2 = 625 & (57-42)^2 &= \\ (15)^2 &= 225 & (113-42)^2 &= (71)^2 = 5,041 & (25-42)^2 &= \\ (-17)^2 &= 289 & 225 &+ 841 &+ 625 &+ 225 &+ 5,041 &+ 289 = 7,246 \end{aligned}$$

To get s⁴, use each variable, subtract the mean, and raise the result to the fourth power. Add all of the results together:

$$(27-42)^4=(-15)^4=50,625(13-42)^4=(-29)^4=707,281(17-42)^4=(-25)^4=390,625(57-42)^4=(15)^4=50,625(113-42)^4=(71)^4=25,411,681(25-42)^4=(-17)^4=83,521$$

$$50,625+707,281+390,625+50,625+25,411,681+83,521=26,694,358$$

So, our sums are:

$$s^2=7,246s^4=26,694,358$$

Now, calculate m_2 and m_4 , the second and fourth moments of the kurtosis formula:

$$m_2=n \cdot s^2=6 \cdot 7,246=1,207.67$$

$$m_4=n \cdot s^4=6 \cdot 26,694,358=4,449,059.67$$

We can now calculate kurtosis using a formula found in many statistics textbooks that assumes a perfectly normal distribution with a kurtosis of zero:

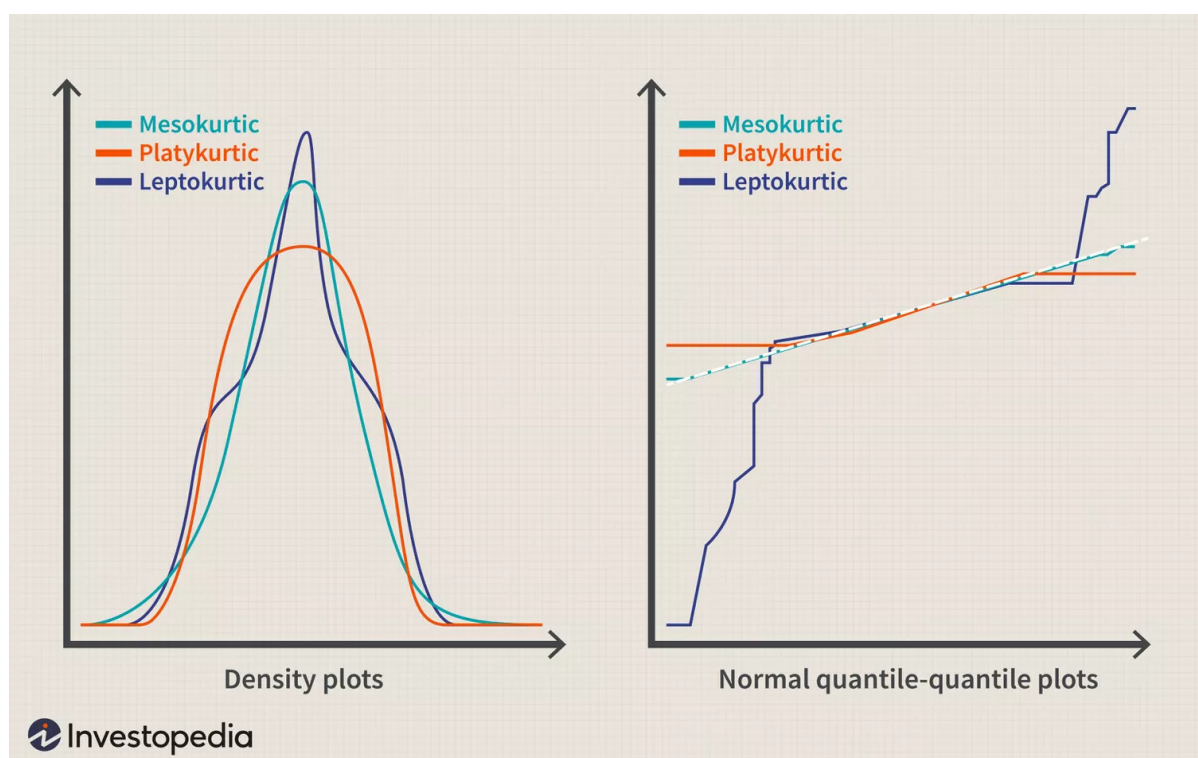
$$k=m_4-3m_2^2 \text{ where: } k=\text{Kurtosis } m_4=\text{Fourth moment } m_2=\text{Second moment}$$

So, the kurtosis for the sample variables is:

$$1,458,466.83-4,449,059.67-3=.05$$

Types of Kurtosis

There are three categories of kurtosis that a set of data can display: mesokurtic, leptokurtic, and platykurtic. All measures of kurtosis are compared against a normal distribution curve.



Mesokurtic (Kurtosis = 3.0)

The first category of kurtosis is mesokurtic distribution. This distribution has a kurtosis similar to that of the normal distribution, meaning the extreme value characteristic of the distribution is similar to that of a normal distribution. Therefore, a stock with a mesokurtic distribution generally depicts a moderate level of risk.

Leptokurtic (Kurtosis > 3.0)

The second category is leptokurtic distribution. Any distribution that is leptokurtic displays greater kurtosis than a mesokurtic distribution. This distribution appears as a curve with long tails (outliers). The “skinniness” of a leptokurtic distribution is a consequence of the outliers, which stretch the horizontal axis of the histogram graph, making the bulk of the data appear in a narrow (“skinny”) vertical range.

A stock with a leptokurtic distribution generally depicts a high level of risk but the possibility of higher returns, because the stock has typically demonstrated large price movements.

While a leptokurtic distribution may be “skinny” in the center, it also features “fat tails.”

Platykurtic (Kurtosis < 3.0)

The final type of distribution is platykurtic distribution. These types of distributions have short tails (fewer outliers). Platykurtic distributions have demonstrated more stability than other curves because extreme price movements rarely occurred in the past. This translates into a less-than-moderate level of risk.

Kurtosis Vs. Skewness

Kurtosis and skewness are both statistical measures used to describe the shape of a probability distribution, but they focus on different aspects. Kurtosis measures the tailedness of a distribution. Skewness measures the asymmetry of a distribution.

Skewness indicates the direction and degree to which the data deviates from a symmetrical bell curve. A distribution with zero skewness is perfectly symmetrical, meaning the left and right sides of the distribution are mirror images. Positive skewness means that the right tail is longer or fatter than the left, suggesting that the data has a tendency to have higher values. Negative skewness indicates that the left tail is longer or fatter, implying a tendency towards lower values.

While skewness focuses on the balance of data around the mean, kurtosis focuses on the distribution's peak and the weight of its tails. A dataset can have high kurtosis with many outliers but still be symmetric and thus have zero skewness. On the other hand, a dataset can be skewed with either positive or negative skewness but has low kurtosis, indicating fewer extreme values.

Using Kurtosis

Kurtosis is used in financial analysis to measure an investment's risk of price volatility. Kurtosis measures the amount of volatility that an investment's price has experienced regularly. High kurtosis of the return distribution implies that an investment will yield occasional extreme returns. Be mindful that this can swing both ways—meaning high kurtosis indicates either large positive returns or extreme negative returns.

For example, imagine a stock had an average price of \$25.85 per share. If the stock's price swung widely and often enough, the bell curve would have heavy tails (high kurtosis). This means that there is a lot of variation in the stock price—an investor should anticipate wide price swings often.

On the other hand, a portfolio with a low kurtosis value indicates a more stable and predictable return profile, which may indicate lower risk. In this light, investors may intentionally seek investments with lower kurtosis values when building safer, less volatile portfolios.

Kurtosis can also be used to strategically implement an investment allocation approach. For example, a portfolio manager who specializes in value investing may prefer to invest in assets with a negative kurtosis value, as a negative kurtosis value indicates a flatter distribution with more frequent small returns. Conversely, a portfolio manager who specializes in momentum investing may prefer to invest in assets with a positive kurtosis value with peaked distributions of less frequent but larger returns.

Kurtosis vs. Other Commonly Used Measurements

Kurtosis risk differs from more commonly used measurements. Alpha measures excess return relative to a benchmark index. While kurtosis measures the nature of the peak or flatness of the distribution, alpha measures the skewness or asymmetry of the distribution.

Beta measures the volatility of a stock compared to the broader market. Each security or investment has a single beta that indicates whether or not that security is more or less volatile compared to a market benchmark. Again, beta measures the degree of asymmetry of the distribution, while kurtosis measures the peak or flatness of the distribution.

R-squared measures the percent of movement a portfolio or fund has that can be explained by a benchmark. Though r-squared is used in regression analysis to assess the goodness of fit of a regression model, kurtosis is used in descriptive statistics to describe the shape of a distribution.

Lastly, the Sharpe ratio compares return to risk. The Sharpe ratio is used by investors to better understand whether the level of returns they are receiving is commensurate with the level of risk incurred. While kurtosis analyzes the distribution of a dataset, the Sharpe ratio more commonly is used to evaluate investment performance.

Why Is Kurtosis Important?

Kurtosis explains how often observations in some datasets fall in the tails vs. the center of a probability distribution. In finance and investing, excess kurtosis is interpreted as a type of risk known as tail risk, or the chance of a loss occurring due to a rare event, as predicted by a probability distribution. If such events are more common than predicted by a distribution, the tails are said to be “fat.”

How Is Kurtosis Used in Finance?

In finance, kurtosis is used to assess the risk of extreme returns in investment portfolios by analyzing the tailedness of return distributions. A higher kurtosis indicates a greater probability of significant deviations from the mean. This means an investment with higher kurtosis is more likely to deviate from its average return.

Is a High Kurtosis Good or Bad?

A higher kurtosis is not inherently good or bad; it depends on the context and the investor's risk tolerance. For instance, high kurtosis indicates more frequent extreme values or outliers, which can imply higher risk and potential for large gains or losses. For some investors, this is good; for others, it's bad.

What Is Excess Kurtosis?

Excess kurtosis compares the kurtosis coefficient with that of a normal distribution. Most normal distributions are assumed to have a kurtosis of three, so excess kurtosis would be more or less than three. However, some models assume that a normal distribution has a kurtosis of zero, so excess kurtosis would be more or less than zero.⁵

Is Kurtosis the Same As Skewness?

No. Kurtosis measures how much of the data in a probability distribution are centered around the middle (mean) vs. the tails. Skewness instead measures the relative symmetry of a distribution around the mean.

The Bottom Line

Kurtosis describes how much of a probability distribution falls in the tails instead of its center. In a normal distribution, the kurtosis is equal to three (or zero in some models). Positive or negative excess kurtosis will then change the shape of the distribution accordingly.

For investors, kurtosis is important in understanding tail risk, or how frequently “infrequent” events occur, given one’s assumption about the distribution of price returns.

